

Buck-Boost (LTC3440) — Working Document

Digital Audio Player · Rev 4

Author: Alexander Loker — design decisions, component selection, analysis and verification.
Drafted by: Claude (Anthropic) — structure and prose from the author's direction and source data.

1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LTC3440EMS#TRPBF
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → 3.9 V intermediate rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 – 4.2 V
Output	3.9 V
Load	Analog chain only (LT3042 +3.3 V; LT3462A inverter → LT3093 –3.3 V)
Output current	≈ 249 mA peak (98 mA LT3042 + 151 mA LT3462A input at 65%); 250 mA design
Fault ceiling	≈ 0.3 A (LT3042 125 mA + LT3462A at its switch limit)
Switching frequency	1.2 MHz, fixed (Burst Mode off)
Output ripple target	0.5% (19.5 mV pp)
Soft-start	Output up by ≈ 6 ms, MCU-enabled
Source resistance	≈ 0.21 Ω

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND
3	SW1	L
4	SW2	L
5	GND	GND
6	VOUT	+3V9, C_OUT, R1
7	VIN	SYS, C_IN
8	SHDN/SS	R_SS from GPIO, C_SS to GND
9	FB	R1 / R2 midpoint
10	VC	C_P1 + R_Z to FB

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L	6.8 μ H $\pm$ 20%	I _{sat} 3.5 A (20%), 74.1 m Ω	Bourns SRP4030FA-6R8M	#1-076
C _{IN}	4.7 μ F	16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
C _{OUT}	22 μ F	25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R ₁	220 k Ω 0.1%	0402	YAGEO RC0402BR-07220KL	#1-074
R ₂	100 k Ω 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
R _T	50 k Ω 0.1%	0603	YAGEO RT0603BRD0750KL	#1-075
R _Z	15 k Ω 0.1%	0402	YAGEO RT0402BRD0715KL	#1-078
C _{P1}	1.5 nF $\pm$ 2%	50 V C0G 0402	Murata GRM1555C1H152GA01D	#1-077
R _{SS}	54.9 k Ω 0.1%	0402	YAGEO RT0402BRD0754K9L	#1-079
C _{SS}	0.1 μ F	50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
R _{PD}	1 M Ω	GPIO side of R _{SS}	—	—

5. Equations and Results

Switching frequency

$$R_T = \frac{6 \times 10^{10}}{f} = \frac{6 \times 10^{10}}{1.2 \times 10^6} = 50 \text{ k}\Omega$$

Output voltage

$$V_{OUT} = 1.22 \left(1 + \frac{R_1}{R_2} \right) = 1.22 \left(1 + \frac{220}{100} \right) = 3.904 \text{ V} \quad (3.827 - 3.981 \text{ V})$$

Inductor ripple (f in MHz; 250 mA design current)

$$Ripple_{boost} = \frac{V_{IN(MIN)}(V_{OUT} - V_{IN(MIN)})}{f I_{OUT} L V_{OUT}} = \frac{3.0 \times 0.9}{1.2 \times 0.25 \times 6.8 \times 3.9} = 0.34 (85 \text{ mA}_{pp})$$

$$Ripple_{buck} = \frac{V_{OUT}(V_{IN(MAX)} - V_{OUT})}{f I_{OUT} L V_{IN(MAX)}} = 0.14 (34 \text{ mA}_{pp})$$

Peak inductor current (3.0 V in, $\eta \approx 0.85$)

$$I_{L,pk} = \frac{0.25 \times 3.9}{3.0 \times 0.85} + \frac{0.085}{2} = 0.42 \text{ A}$$

Output capacitance (0.5% ripple)

$$C_{OUT} \geq \frac{I_{OUT}(V_{OUT} - V_{IN(MIN)}) \cdot 100}{\%Ripple \cdot V_{OUT}^2 \cdot f} = \frac{0.25 \times 0.9 \times 100}{0.5 \times 3.9^2 \times 1.2 \times 10^6} = 2.47 \mu F$$

Soft-start

$$\tau = R_{SS}C_{SS} = 5.49 \text{ ms}, \quad t_x = \tau \ln \frac{V_{GPIO}}{V_{GPIO} - V_x}$$

Enable (1.0 V): 2.0 ms · Ramp complete (2.2 V): 6.0 ms

Four-switch region

$$V_{IN} = V_{OUT}(1 - 150 \text{ ns} \cdot f) \text{ to } \frac{V_{OUT}}{1 - 150 \text{ ns} \cdot f} = 3.20 \text{ to } 4.76 \text{ V}$$

Load (0.6 V RMS max output; LDO limits 125 mA each)

Case	LT3042	LT3462A input (η 65%)	LTC3440 load
Quiescent	27 mA	$\approx 42 \text{ mA}$	$\approx 69 \text{ mA}$
Full scale, 23.8 Ω	98 mA	$\approx 151 \text{ mA}$	$\approx 249 \text{ mA}$
Fault (LDOs / inverter in limit)	125 mA	$\approx 0.18 \text{ A}$	$\approx 0.3 \text{ A}$

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 - 2.9 A)
Simulated output ripple	7.7 mV pp (target 19.5 mV)
Load step	150 $\leftrightarrow$ 250 mA $\pm$ 90 mV, no ringing, 0.3 ms recovery (conservative vs 196 mA peak)
Loop crossover (implied)	$\approx 8 \text{ kHz}$
Minimum rail during step	3.74 V
ESL spikes (0.5 nH, sim)	$\approx 0.3 \text{ V}$ — layout item
Peak load	$\approx 249 \text{ mA}$ (at design current)
Fault ceiling	$\approx 0.3 \text{ A}$ — inductor peak $\approx 0.5 \text{ A}$ vs 3.5 A I_{sat} , switch 2.5 A min

7. Open Problems

#	Problem
1	LT3462A efficiency assumed 65% — peak load sits at the 250 mA design current; confirm in LTspice and re-check ripple and load step

8. Revision Log

Rev	Date	Change
0	2026-09-21	Initial design, calculations and simulation.
1	2026-09-21	Schematic captured; VC-to-FB short corrected; MPN to #TRPBF.
2	2026-09-21	Condensed to essentials.
3	2026-09-22	Load updated to 0.6 V RMS max output (≈ 196 mA peak) and 250 mA fault ceiling from 125 mA LDO current limits.
4	2026-09-23	Negative branch now LT3462A inverter: peak load ≈ 249 mA, fault ≈ 0.3 A.